

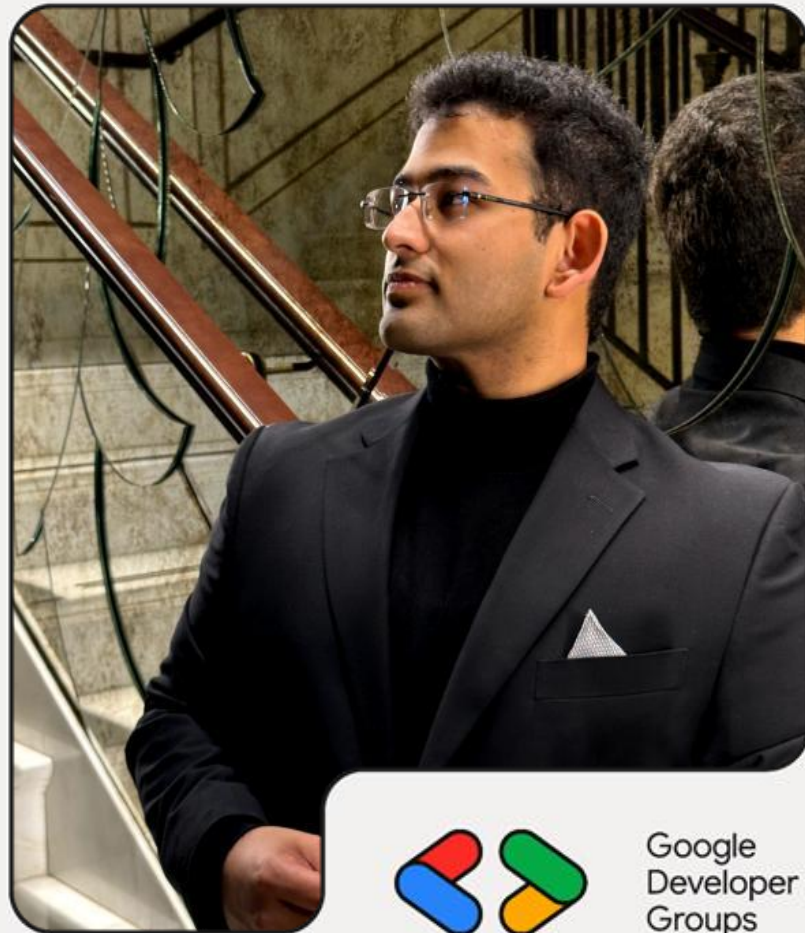


Dubai, UAE

Building And Deploying An Agentic AI App With Google AI Studio And ADK

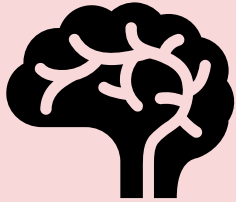
Rohan Mitra

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ML Engineer @ Bayut | Dubizzle
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LLMs vs Agentic AI



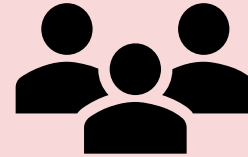
LLM

- Reasoning and knowledge, but cant take actions
- Stateless
- Passive



Agent

- Body = tools
- Job Desc = instructions & goals
- Autonomous
- Stateful
- Take actions till job is done



Multi-Agent

- Talented and focused individuals
- Get tasks to help achieve goal
- Focused
- Modular, reliable, debug-able

Why ADK?

- Need control to build complex agents and workflows
 - Modularity
 - Reusability
 - State management
- Easy deployment and integration
- Quick set up while being scalable

Case Study – Research Explainer

- Take a research paper / technical document and explain it while generating required diagrams and charts
- One Agent to read and understand the paper
- Tool call to generate flow charts – show processes
- Tool call to find external research context for a concept discussed – show applications

Our Architecture

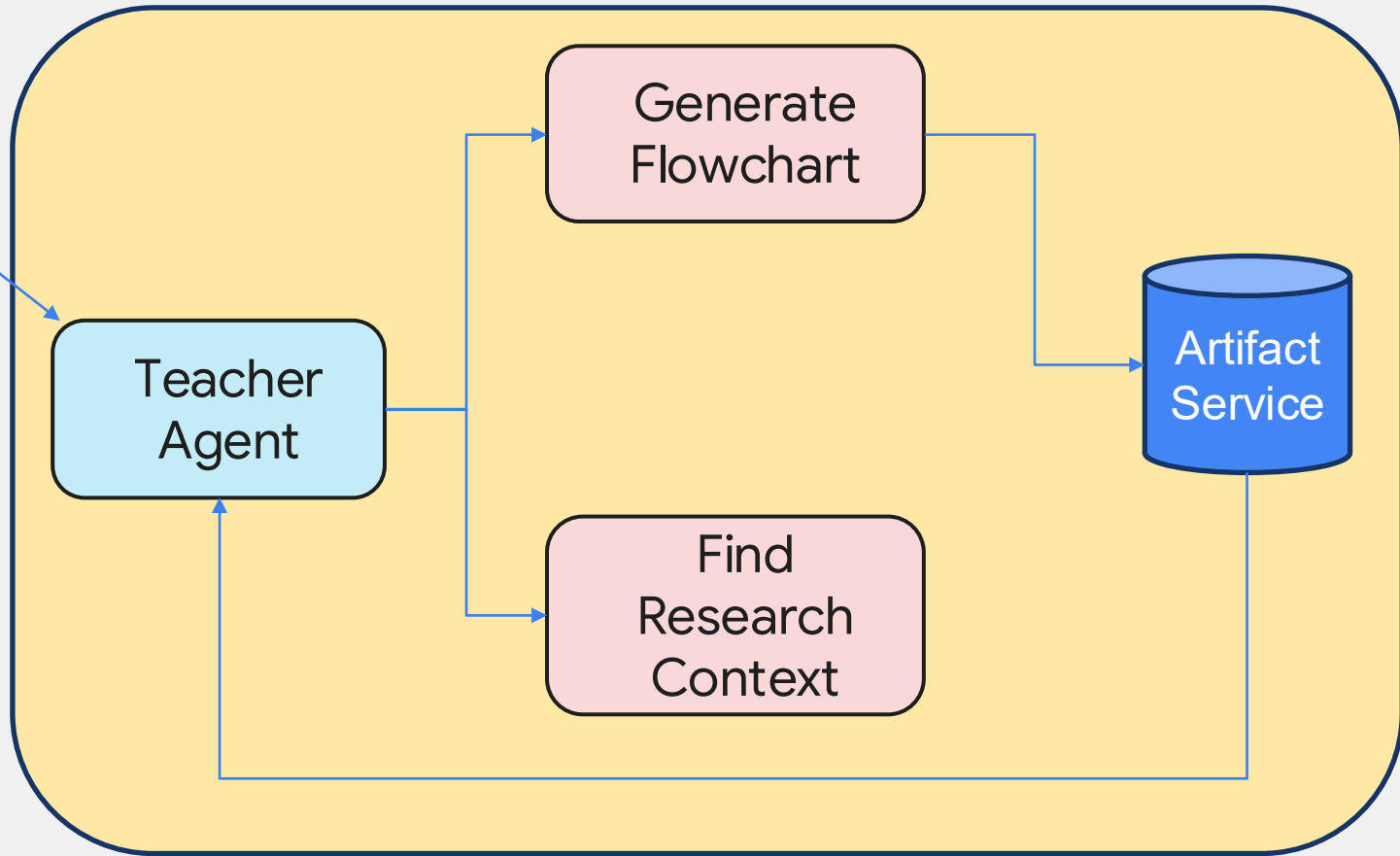
Interactive
Website

Teacher
Agent

Generate
Flowchart

Find
Research
Context

Artifact
Service



Our Deployment Strategy

Frontend

Interactive
Website



Firebase

Backend

ADK Agent



Cloud Run

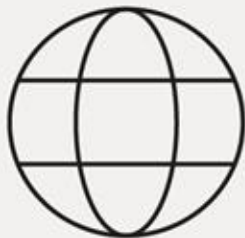
Artifact & Sessions

Artifacts

- Any large object that is loaded/created
- Includes files, images, data, etc
- **Artifact service:** A temporary storage space for the session
- Easy way to transfer large objects between agents
- Saves to RAM / GCS / Redis

Sessions

- Agent's memory of this conversation
- Handled by ADK
- **Session service:** Where the session details will be saved (temp)
- Important when multiple agent pods exist – share the same memory
- Saves to RAM / GCS / Redis



Demo Code on Github

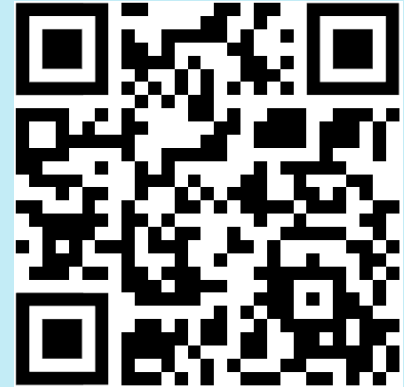
github.com/ro1406/research-paper-explainer



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Deployment Options

- Deploy directly to GCP – Cloud Run
- Non-GCP deployments – Docker container
 - Single service
 - Host as ``adk api_server -port 8000``
- Identify good Artifact services
 - RAM – local / low traffic / single deployment
 - GCS good for GCP deployments
 - Redis sidecar for kubernetes deployments
- Serverless deployments
 - Event driven
 - Scalable (to 0 if needed)



Medium.com Article

Deployment - Setup

- Install GCloud:
`https://docs.cloud.google.com/sdk/docs/install-sdk`
- Install Firebase CLI: `npm install -g firebase-tools`
- Create GCP Project
- Enable VertexAI API (Skip if you're using AI Studio API Key)

New Project

 You have 17 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[Manage Quotas](#)

Project name *
My Project 44871 

Project ID: affable-hydra-499620-h7. It cannot be changed later. [Edit](#)

Billing account *
Google Cloud Platform Trial Billing Account

Any charges for this project will be billed to the account you select here.

Parent resource *
 No organization [Browse](#)

Select the organization or folder where this resource will be created

[Create](#) [Cancel](#)

 Gemini Enterprise

Welcome to Agent Platform

Vertex AI is now Agent Platform. Build enterprise-grade agents with the latest models.

Get started with Generative AI

Access multimodal capabilities to generate text, code, and images with Gemini or your chosen model.

[Enable APIs](#) [View setup docs](#)

Deployment – Link Firebase

- Go to `console.firebase.google.com`
- Click **“Add Project”** -> Choose **“Add Firebase to a Google Cloud Project”**
- Select existing GCP Project
- Follow the prompts

Enter your project name

my-awesome-project-id

Already have a Google Cloud project?
[Add Firebase to Google Cloud project](#)

Continue

Deployment – Local GCP Setup

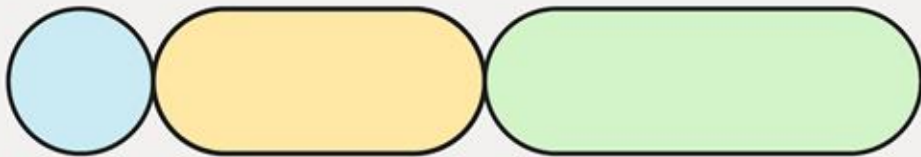
- In local console, run `gcloud auth login`
- Run `gcloud init` and set project name
- Export the required env vars
 - `GOOGLE_CLOUD_LOCATION`
 - `GOOGLE_GENAI_USE_VERTEXAI` (if using VertexAI API, else `GOOGLE_API_KEY`)
 - `GOOGLE_CLOUD_PROJECT_ID`
- Set project name in `.firebase`rc
- Run `firebase login`

Deployment – Cloud Run Deploy

```
gcloud run deploy research-explainer \  
  --source . \  
  --region us-central1 \  
  --min-instances=1 \  
  --max-instances=1 \  
  --allow-unauthenticated \  
  --set-env-vars="GOOGLE_API_KEY=<your_api_key>"
```

Deployment – Firebase Deploy

- Set the URL from cloud run deploy into `index.html` `BACKEND_URL`
- `firebase deploy --only hosting`



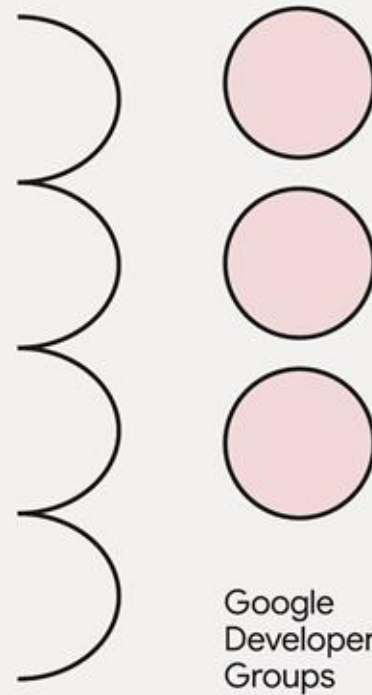
Thank You!



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